# Advanced Plane Geometry

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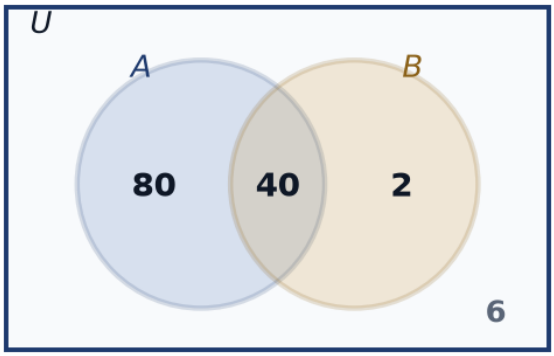
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Circle theorems: tangent-chord and secant angles

Two secants are drawn from an external point to a circle, intercepting arcs of $100^\circ$ and $30^\circ$. Find the measure of the angle formed at the external point, using the formula (difference of arcs) ÷ 2.

**2** [Foundational] • Target: Circle theorems: tangent-chord and secant angles

Two chords intersect inside a circle, intercepting arcs of $80^\circ$ and $40^\circ$. Find the measure of the angle formed at the intersection, using the formula (sum of arcs) ÷ 2.




**3** [Foundational] • Target: Power of a point (secant and tangent segment relationships)

A tangent segment from an external point has length 8, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

**4** [Foundational] • Target: Area of regular polygons

Find the area of a regular pentagon with side length 6 and apothem 4.13, using $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem}$.

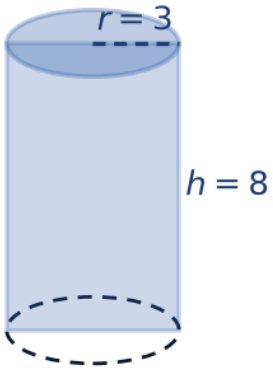
**5** [Foundational] • Target: Similar solids and scale factor in 3D

Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 40, find the volume of the larger cylinder.

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**6** [Foundational] • Target: Composite 3D solids

A solid consists of a cylinder of radius 3 and height 8, topped with a hemisphere of the same radius. Find the total volume, in terms of $\pi$.




**7** [Foundational] • Target: Triangle centers: centroid and circumcenter

Find the centroid of the triangle with vertices $(2, 3)$, $(8, 3)$, $(5, 9)$, using the formula (average of the vertex coordinates).

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Heron's formula

A triangle has sides 7, 8, and 9. Find its area using Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s$ is the semi-perimeter.

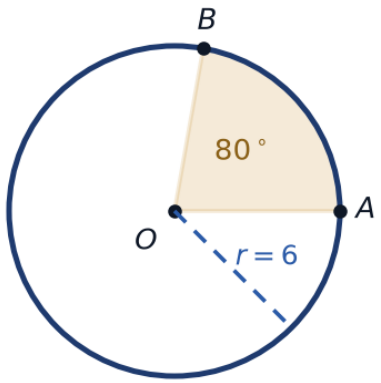
**9** [Intermediate] • Target: A multi-part circle geometry problem

This question has two parts. A circle has radius 10. A chord is drawn 6 units from the center. (a) Find the length of the chord, using the relationship between the radius, the distance from center to chord, and half the chord length (a right triangle relationship). (b) Find the area of the minor segment cut off by this chord, to the nearest whole number, given that the central angle subtended by the chord is approximately $106.3^\circ$ and $\sin(106.3^\circ) \approx 0.960$.

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**10** **[Intermediate]** • **Target: Angle bisectors and the angle bisector theorem**

In triangle $ABC$, the angle bisector from $A$ meets side $BC$ at point $D$, dividing it such that $BD = 6$ and $DC = 9$. If $AB = 8$, find $AC$, using the angle bisector theorem ($\frac{BD}{DC} = \frac{AB}{AC}$).




**11** **[Intermediate]** • **Target: The 30-60-90 and 45-45-90 triangles in composite figures**

An equilateral triangle has side length 10. Find its height and area, using the fact that the height splits it into two 30-60-90 triangles.

**12** **[Intermediate]** • **Target: Circumscribed and inscribed circles**

A square is inscribed in a circle of radius 5. Find the side length of the square, using the fact that the diagonal of the square equals the diameter of the circle.

**13** **[Intermediate]** • **Target: Regular polygon interior and exterior angles**

Find the measure of each interior angle and each exterior angle of a regular octagon.

**14** **[Intermediate]** • **Target: A multi-part composite figure and circle problem**

This question has three parts. A running track consists of a rectangle with two semicircular ends. The rectangle is 80 meters long and 40 meters wide, and the semicircles are attached to the two 40-meter-wide ends. (a) Find the radius of each semicircle. (b) Find the total perimeter (distance around) the track, in terms of $\pi$. (c) Find the total enclosed area of the track shape, in terms of $\pi$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

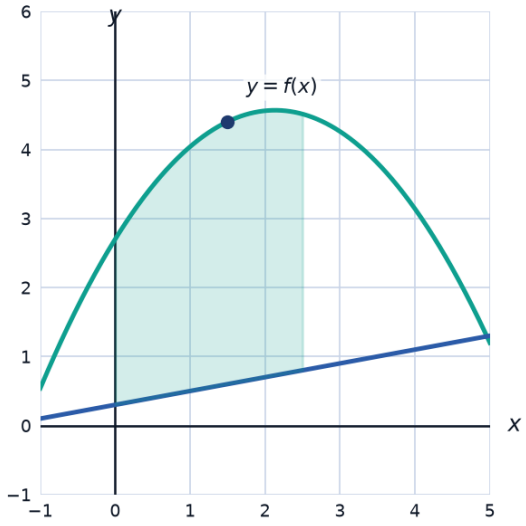
**15** **[Challenging]** • **Target: Power of a point (secant and tangent segment relationships)** • **Cross-Lesson: 8.01 Statistics, Counting & Probability**

Two chords $AB$ and $CD$ intersect inside a circle at point $X$. If $AX = 4$ cm, $XB = 9$ cm, and $CX = 3$ cm, calculate the length of segment $XD$ and the total length of chord $CD$.

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**16** [Challenging] • Target: Area of regular polygons • Cross-Lesson: 1.01 Advanced Number Theory

A regular hexagon has a side length of $s = 8$ cm. Find the length of its apothem $a$ and its exact total area $A$.




**17** [Challenging] • Target: Similar solids and scale factor in 3D • Cross-Lesson: 2.01 Advanced Algebra & Systems

A right circular cone has a base radius of 6 cm and a slant height of 10 cm. Find (a) its perpendicular vertical height $h$, (b) its volume $V$ in terms of $\pi$, and (c) its total surface area in terms of $\pi$.

**18** [Challenging] • Target: Composite 3D solids • Cross-Lesson: 3.01 Advanced Functions

A tangent from an external point $P$ touches a circle at $T$. If the distance from $P$ to the circle's centre $O$ is 13 and the radius is 5, find the length of the tangent segment $PT$.

**19** [Challenging] • Target: Triangle centers: centroid and circumcenter • Cross-Lesson: 4.01 Advanced Trigonometry

A triangle has side lengths $a = 5$, $b = 6$, and $c = 7$. Use Heron's formula to compute its exact area.

**20** [Challenging] • Target: Heron's formula • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the sum of interior angles of a polygon with 9 sides.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Circle theorems: tangent-chord and secant angles

Find the area of a rectangle with length 19 and width 18.

**a.** 74

**b.** 342

**c.** 360

**d.** 323

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**22** [Foundational] • Target: Power of a point (secant and tangent segment relationships)

Find the measure of each interior angle of a regular polygon with $18$ sides.

**a.** 169

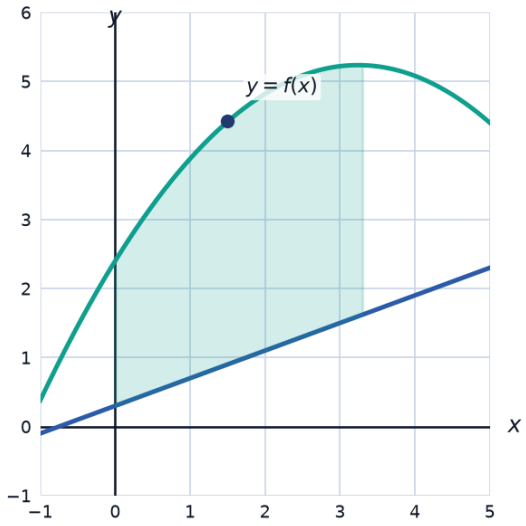
**b.** 163

**c.** 162

**d.** 160

**23** [Foundational] • Target: Area of regular polygons

Two similar solids have a linear scale factor of $2:1$. If the surface area of the smaller solid is $12$, find the surface area of the larger solid.




**a.** 14

**b.** 2

**c.** 13

**d.** 3

**24** [Foundational] • Target: Similar solids and scale factor in 3D

Find the measure of each exterior angle of a regular polygon with $12$ sides.

**a.** 36

**b.** 30

**c.** 32

**d.** 21

**25** [Foundational] • Target: Composite 3D solids

A regular polygon has an exterior angle of $36^\circ$. How many sides does it have?

**a.** 6

**b.** 10

**c.** 7

**d.** 4

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**26** [Foundational] • Target: Triangle centers: centroid and circumcenter

Find the perimeter of a rectangle with length 9 and width 20.

**a.** 180

**b.** 60

**c.** 29

**d.** 58

**27** [Foundational] • Target: Heron's formula

In triangle $ABC$, point $D$ lies on side $BC$ such that $AD$ is an angle bisector of angle $A$. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

**a.** This statement is a direct consequence of the Sine Rule applied to triangles $ABD$ and $ACD$.

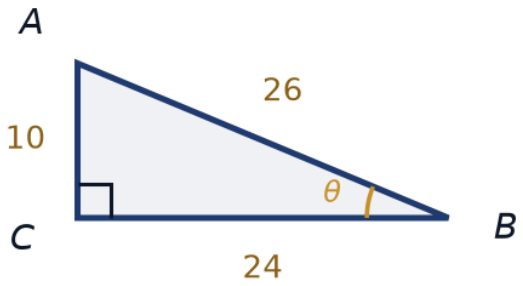
**b.** This statement can be proven by applying the Law of Cosines to triangles $ABD$ and $ACD$ and equating expressions for $AD^2$.

**c.** This statement is true only if triangle $ABC$ is isosceles with $b = c$.

**d.** This statement requires the use of vector methods for its proof.

**28** [Foundational] • Target: A multi-part circle geometry problem

In the diagram below, two similar triangles ABC and ADE are shown, with point D on segment AB and point E on segment AC. If AD = 4, DB = 8, and AE = 6, what is the length of EC?




**a.** 12

**b.** 10

**c.** 8

**d.** 9

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** **[Intermediate]** • **Target: Angle bisectors and the angle bisector theorem**

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. An altitude is drawn from C to the hypotenuse AB, meeting AB at point D. If AC = 9 and BC = 12, find the length of the altitude CD.

**a.** 7.2

**b.** 10.8

**c.** 8.4

**d.** 9.6

**30** **[Intermediate]** • **Target: The 30-60-90 and 45-45-90 triangles in composite figures**

The three angles of a triangle measure $(3x - 5)^\circ$, $(3x - 10)^\circ$, and $(x + 20)^\circ$. Find the measure of the largest angle.

**a.** 70

**b.** 64

**c.** 62

**d.** 65

**31** **[Intermediate]** • **Target: Circumscribed and inscribed circles**

Two similar solids have a linear scale factor of $1:4$. If the volume of the smaller solid is $13$, find the volume of the larger solid.

**a.** 443

**b.** 655

**c.** 832

**d.** 193

**32** **[Intermediate]** • **Target: Regular polygon interior and exterior angles**

What is the sum of the interior angles of a convex polygon with $4$ sides?

**a.** 367

**b.** 360

**c.** 368

**d.** 358

**33** **[Intermediate]** • **Target: A multi-part composite figure and circle problem**

The sum of the interior angles of a convex polygon is $2520^\circ$. How many sides does the polygon have?

**a.** 20

**b.** 16

**c.** 19

**d.** 3

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**34** **[Intermediate]** • **Target: Circle theorems: tangent-chord and secant angles**

Two similar solids have a volume ratio of $125:1$. Find the linear (side-length) scale factor.

**a.** 1:9

**b.** 1:10

**c.** 1:5

**d.** 1:7

**35** **[Intermediate]** • **Target: Power of a point (secant and tangent segment relationships)**

Three distinct vertices of a regular 12-gon are selected uniformly at random. What is the probability that the triangle formed by these three vertices is obtuse?

**a.** $\frac{3}{11}$

**b.** $\frac{5}{11}$

**c.** $\frac{6}{11}$

**d.** $\frac{7}{11}$

**36** **[Intermediate]** • **Target: Area of regular polygons**

In $\Delta ABC$, the side lengths opposite angles $A, B,$ and $C$ are $a, b,$ and $c$, respectively. Given that $a = 7$, $b = 8$, and $\sin(B - A) = \frac{3}{5}\sin C,$ what is the length of side $c$?

**a.** $5$

**b.** $3\sqrt{5}$

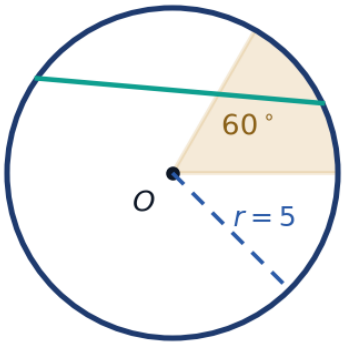
**c.** $\frac{5}{3}$

**d.** $2\sqrt{7}$

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**37** **[Intermediate]** • **Target: Similar solids and scale factor in 3D**

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?




**a.** 6 units

**b.** $6\sqrt{3}$ units

**c.** 12 units

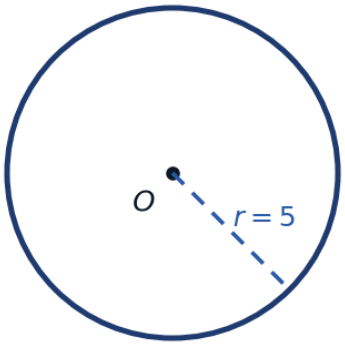
**d.** $12\sqrt{3}$ units

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Composite 3D solids • Cross-Lesson: 3.01 Advanced Functions

On the unit circle, a point $P$ is located at coordinates $(x, y)$. If the angle formed by the positive x-axis and the line segment from the origin to $P$ is $\theta$, what is the value of $x$ in terms of $\theta$?




**a.** $\cos\theta$

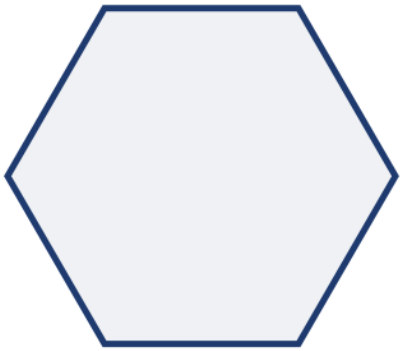
**b.** $\sin\theta$

**c.** $\tan\theta$

**d.** $1$

**39** [Challenging] • Target: Triangle centers: centroid and circumcenter • Cross-Lesson: 4.01 Advanced Trigonometry

The sum of interior angles of a hexagon is:



**a.** 540°

**b.** 720°

**c.** 900°

**d.** 360°

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**40** [Challenging] • Target: Heron's formula • Cross-Lesson: 5.01 Pre-Calculus Concepts

A circle has equation $(x-1)^2+(y+3)^2=9$. What are its center and radius?

**a.** Center $(-1, 3)$, radius $9$

**b.** Center $(-1, 3)$, radius $3$

**c.** Center $(1, -3)$, radius $9$

**d.** Center $(1, -3)$, radius $3$

**41** [Challenging] • Target: A multi-part circle geometry problem • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Parallel lines have slopes that are:

**a.** Negative reciprocals

**b.** Equal

**c.** Opposite signs

**d.** Unrelated

**42** [Challenging] • Target: Angle bisectors and the angle bisector theorem • Cross-Lesson: 8.01 Statistics, Counting & Probability

In the diagram below, line $AB$ is parallel to line $CD$. Transversal line $EF$ intersects $AB$ at point $G$ and $CD$ at point $H$. If the measure of angle $AGH$ is $50^\circ$, what is the measure of angle $GHD$?

**a.** 50°

**b.** 130°

**c.** 40°

**d.** 90°

**43** [Challenging] • Target: The 30-60-90 and 45-45-90 triangles in composite figures • Cross-Lesson: 1.01 Advanced Number Theory

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is $\theta$. If $\sin(\theta)=\frac{3}{5}$ and the y-coordinate of $P$ is $y=\frac{3}{5}$, determine the x-coordinate of $P$.

**a.** $-\frac{4}{5}$

**b.** $\frac{4}{5}$

**c.** $-\frac{3}{5}$

**d.** $-\frac{25}{16}$

**44** [Challenging] • Target: Circumscribed and inscribed circles • Cross-Lesson: 2.01 Advanced Algebra & Systems

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are $A$ and $B$. Determine the length of the chord $AB$.

**a.** 6

**b.** 8

**c.** 10

**d.** 4

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**45** [Challenging] • Target: Regular polygon interior and exterior angles • Cross-Lesson: 3.01 Advanced Functions

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. CD is the altitude to the hypotenuse AB. If AD = 4 units and DB = 9 units, find the length of the altitude CD.

**a.** 6 units

**b.** 13 units

**c.** 36 units

**d.** 7.5 units

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